UNIVERSITI TEKNOLOGI MALAYSIA

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SECV2113 HUMAN COMPUTER INTERACTION

SECTION 01

ASSIGNMENT 2

HEURISTIC EVALUATION & USER JOURNEY MAPPING

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GROUP 9

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Introduction

In Assignment 2, we will evaluate Group 3 prototype using Heuristic Evaluation (HE). In this evaluation, we apply Nielsen's heuristics and rate the severity of each issue found.

10 rules in Nielsen's Heuristics

1. **Visibility of system status**
   Keep users informed about system status with timely feedback

2. **Match between system and real world**
   Use users familiar language and concepts, presenting information logically

3. **User control and freedom**
   Ask for confirmation before performing actions, allow undo, redo and cancel action

4. **Consistency and standards**
   Users interact without having to wonder how to do it or learn new actions

5. **Error prevention**
   Design the system to prevent errors before they occur

6. **Recognition rather than recall**
   Users recognize interface elements without the need to analyze them in detail each time

7. **Flexibility and efficiency of use**
   Provide the user with different ways to get to the content, accelerators and custom actions

8. **Aesthetic and minimalist design**
   Keep the design simple, avoiding unnecessary elements

9. **Help users recognize, diagnose, and recover from errors**
   Provide clear error messages and solutions

10. **Help and documentation**
    Provide help and documentation for the user

Severity Rating

1. Cosmetic Issue
2. Minor Issue
3. Major Issue

4. Catastrophic issue

Evaluation Process

After receiving detailed user personas and scenarios from the group, we started our evaluation process by reading them carefully. First, we tested the prototype together to find major usability problems. We recorded these initial issues in a results table, noting which heuristics they violated. Next, each of us individually reviewed the application using Nielsen's heuristics, finding more issues and rating their severity. We then met as a group to discuss our findings, share opinions, and agree on the most important issues. Finally, we compiled the agreed-upon issues and their severity ratings into a final report.

Prototype Overview

Group 3's prototype consists of many features that are provided to the user.

1. An integrated healthcare record management system for consolidating health information and providing personalized insights.
2. An online chatting platform with AI characters for mental health support and professional consultations.
3. Features like free online consultations, appointment reminders, and tailored healthcare services for enhanced user convenience.

Group 3 Figma:
https://www.figma.com/design/F7S3dAXtpJGrI3ZPEDZdak/Healthcare-App-with-Blush-Illustrations-(Community)?node-id=6-0&t=gQ4rGa7NHqfSg46u-1

User Persona & Scenario

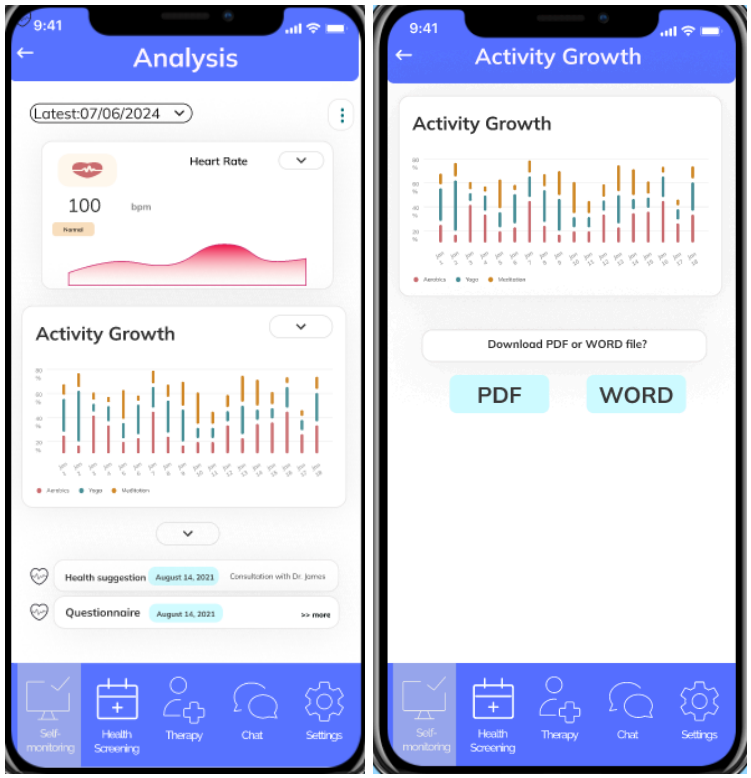
We evaluated their prototype using Ms. Kong's persona, a housewife and business owner. More details are explained in the User Journey Maps Section A.

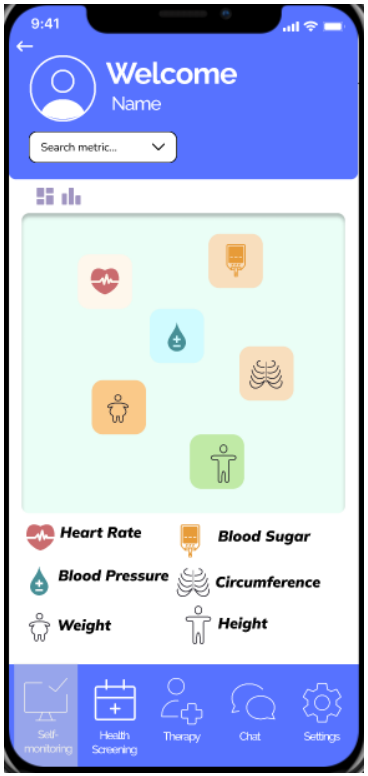
Her main tasks are to:

- Track daily health data efficiently
- Book medical checkup appointments easily and straightforwardly
- Obtain mental health support in her busy life

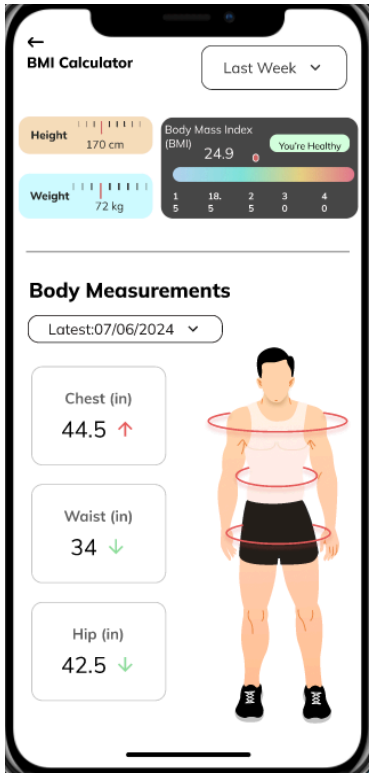
One day, Ms. Kong faced a heart rate issue and decided to track her daily health. However, when she tried to book an appointment, it was full, and she felt anxious. She needed to find someone to listen to her in order to reduce her anxiety.

Heuristic Evaluation

No	Prototype Image	Identified Issue	Heuristic and Severity
1		The “arrow going down” button next to activity growth is confusing, as it appears to lead to more details but actually initiates printing the analysis.	H3: User control and freedom S3: Major issue

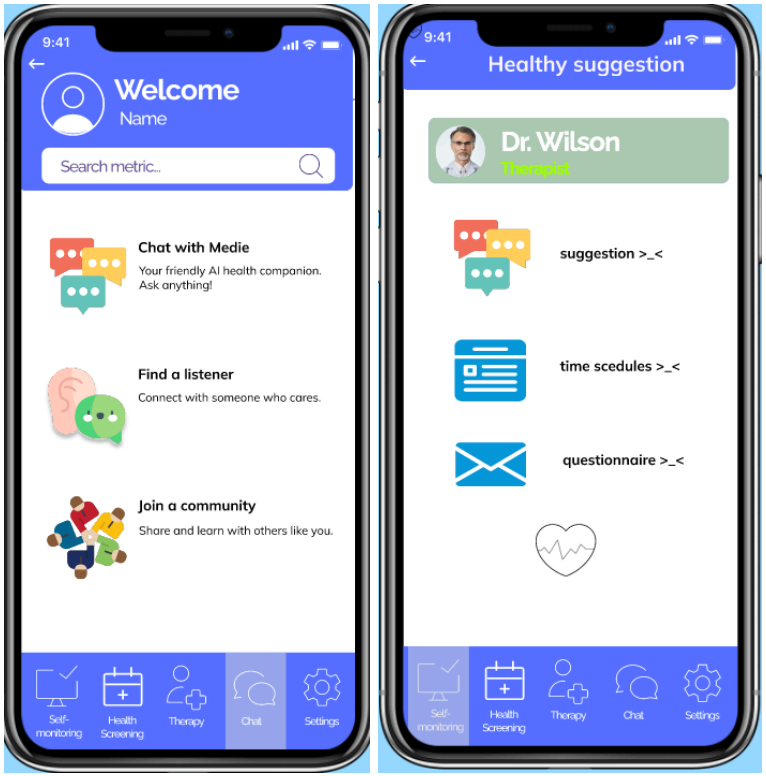
No	Prototype Image	Identified Issue	Heuristic and Severity
2		The center part of the interface is visually cluttered and appears messy and random, with icons not grouped in a logical or organized manner.	H8: Aesthetic and minimalist design S2: Minor issue

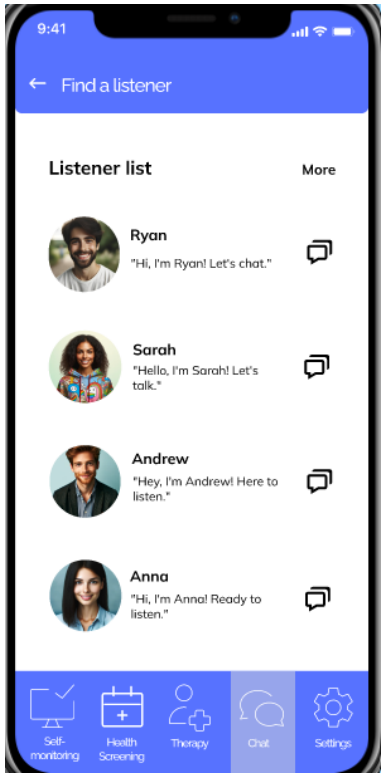
No	Prototype Image	Identified Issue	Heuristic and Severity
3		The "Analysis" button at the main page leads to the following page which is the heart rate analysis. However, the same page also appears when the user selects "Self-monitoring" >> "Heart Rate". The user may question why the heart rate page is shown for the analysis and not the other metrics. Besides, the analysis is also redundant.	H4: Consistency and Standards S3: Major Issue

No	Prototype Image	Identified Issue	Heuristic and Severity
4		For the analysis page of each metric, there is a menu-like button on the top right corner which leads to the following BMI calculator. It confuses the users about the relationship of each metrics, eg. blood sugar, heart rate and blood pressure, to BMI calculations.	H2: Match between system and real world. S2: Minor issue

No	Prototype Image	Identified Issue	Heuristic and Severity
5		The "More" icon brings the users to a page that shows more information about the doctor, along with the option to get an appointment. Initially, I assumed that the "More" button solely provided doctor information. This initial assumption suggested that the user can only check appointments and booking appointments was not possible on this page.	H6: Recognition rather than recall S4: Catastrophic issue

No	Prototype Image	Identified Issue	Heuristic and Severity
6		In the main page of the app, it shows the search bar and “Show All” which attracts me to click on it but it shows nothing. It proves that the search bar and “Show All” is not necessary as there is nothing for the user to search because all functions have been displayed in the main page.	H2: Match between system and real world H8: Aesthetic and minimalist design S2: Minor Issue

No	Prototype Image	Identified Issue	Heuristic and Severity
7		The icon for chat with Medie and suggestion is quite similar which may lead to user confusion while using the app.	H4: Consistency and standards S1: Cosmetic Issue

No	Prototype Image	Identified Issue	Heuristic and Severity
8		Lack of listener descriptions requires users to recall previous interactions or guess which listener might be suitable based on limited information.	H6: Recognition rather than recall S3: Major issue

No	Prototype Image	Identified Issue	Heuristic and Severity
9		In the main page, they listed out the top doctors but when I wish to click on their profile to know more, there is no way to go. Presumably, there won't be only these two doctors displayed at the main page, and if the user cannot make an appointment thru this top doctors ranking, there is no need to put it here as the user will not remember the doctor's name when they make an appointment later on.	H6: Recognition rather than recall H8: Aesthetic and minimalist design S2: Minor issue

User Journey Map

Section A – The provided persona + scenario	**Persona**: Ms. Kong's, a housewife and business owner who focused on maintaining her health while managing family and business duties. Her main goals are to track daily health data efficiently, book medical checkup appointments easily, and access mental health support in her busy life conveniently. **Scenario**: She decides to track her daily health data due to her hectic schedule. One day, she notices an issue with her heart rate and becomes concerned. She immediately tries to book a medical checkup appointment through the app, but finds that all available times are fully booked, with the earliest appointment two weeks away. During the waiting period, she feels anxious and needs someone to talk to about her concerns. Then, Mrs. Kong tries to find a listener who can help her by using this system. However, she struggles to find a suitable listener due to limited information provided about their backgrounds and experiences. Despite these challenges, she persists in finding someone to talk to and hopes to alleviate her anxiety while waiting for her medical appointment.		
Section B – Journey phases based on the user tasks	Task 1: Track daily health data	Task 2: Book medical checkup appointment	Task 3: Obtain mental health support in her busy life
Section C – List the actions taken by the user in each phase	1. Click on the "Self-monitoring" icon 2. Select the metrics that I would like to monitor. 3. The analysis of the health data will be displayed	1. Click on the "Therapy" icon. 2. Click the "More" icon in the online therapy section. 3. View the available periods. 4. Click on "Get Appointment". 5. Select the date and time. 6. Click "Book Session". 7. A "Booking Session Completed" message will pop up.	1. Click the "Chat" icon 2. Select "Find a listener" 3. Choose a listener from the list 4. Click the message icon to start chatting

Section D – Emotion plots and quotes on the user's thoughts and feelings in each phase	1. Easily find the "Self-monitoring" icon at the home page 2. Various metrics are easily to navigate to	1. Easily find the "Therapy" icon. 2. Click all the icons on the page because it's unclear where to book an appointment. 3. Only the date is given, but no time is specified. 4. Feel relieved upon seeing the appointment confirmation message.	1. Most of the actions are easy to perform and clearly visible. 2. Difficult in finding a suitable listener because they lack descriptions, making it hard to know who has the right personality or experience to help
Section E – Highlight the issues found during heuristic evaluation as pain points. Identify the opportunities (provide recommendations for improvements).	**Redundant Button Usage** **Issue:** The "Analysis" button leads to the heart rate analysis, which can be achieved through the "Self-monitoring" button and then the "Heart Rate" selection. The "Analysis" button seems to be redundant. **Improvement:** Remove the "Analysis" button	**Inconsistent Icon Usage** **Issue**: The "More" icon in the online therapy section confuses users, who think it provides additional information instead of booking options. **Improvement**: Redesign the "More" icon to clearly indicate its function, such as "Book Here." **Incomplete Information** **Issue**: Lack of time information can lead to confusion and scheduling difficulties. **Improvement**: Display both date and time for available periods, ensuring all scheduling information is clearly visible.	**Incomplete Information** **Issue**: Insufficient information about listeners makes it hard for users to find the right one. **Improvement**: Add listener descriptions and matching tags to highlight their expertise and experiences, making it easier for users to find the right listener.